Knowledge before Industrialization:

Evidence from Early Copyright Records

Tancredi Rapone*

Universitat Autònoma de Barcelona and BSE

October 2026

Abstract

This paper examines whether the geography of intellectual production anticipated American industrialization. I assemble copyright registrations for 1790–1870 and link authors active in 1840–1860 to their counties of residence in the 1850 census. Registrations per capita began a sustained rise around 1830, before the major industrial expansion of the late nineteenth century. Counties with more early copyrighting subsequently developed larger manufacturing employment shares and higher patenting rates, even after controlling for geography, literacy, schooling, and initial population and settlement characteristics. The manufacturing relationship is strongest around 1920 and fades later in the twentieth century, whereas the association with population density persists. A migration-based instrument combines county birthplace shares with authorship rates in migrants' places of origin; the resulting estimates are also positive for manufacturing and invention. Finally, the relationship is not confined to scientific or academic subjects: non-academic copyrighting also predicts manufacturing specialization. These findings place a broad range of written intellectual production among the early characteristics of the places where American industry subsequently developed.

Keywords: Knowledge production; copyright; human capital; industrialization; U.S. economic history.

JEL classification: N31, N61, O14, O31.

*I thank Sophie Brochet, Alejandro Gobierno, Jacob Hess, Andre Groeger, and Raül Santaeulàlia-Llopis for helpful comments and advice. I acknowledge support from Universitat Autònoma de Barcelona and the Barcelona School of Economics. All errors are my own.

1 Introduction

The accumulation and application of knowledge lie at the heart of sustained economic growth (Mokyr, 2002). New ideas can improve production, while their preservation and circulation allow further advances to build on what is already known (Mokyr, 2009). Understanding the origins of industrialization therefore requires examining the intellectual activity behind technological change: where knowledge was produced, what subjects it addressed, and how its development related to the transformation of the economy.

Yet knowledge is difficult to observe in historical data. Schooling and literacy describe people's capacity to acquire and use knowledge, while patents record particular inventions. These measures tell us less about the broader range of written work through which ideas were articulated and communicated, or how the subjects of that writing changed over time. Copyright records offer a complementary perspective: they identify registered works and their authors, allowing us to study the production and content of writing during economic development.

This paper studies the relationship between early intellectual production and the geography of American industrialization. I assemble copyright title pages preserved by the Library of Congress for 1790–1870 and link authors of works registered between 1840 and 1860 to their residences in the 1850 population census. The resulting data locate writers across U.S. counties before the large-scale industrial expansion of the late nineteenth century. They also allow me to distinguish writing associated with academic and instructional subjects from other registered works.

Two findings emerge. First, copyright registrations per capita began a sustained rise around 1830, before the larger industrial transformation and the subsequent acceleration in patenting. Written production was expanding well before the period usually associated with American "knowledge-based progress" (Abramovitz and David, 2000). Second, counties with more copyright activity around 1850 subsequently became more specialized in manufacturing and recorded more patents per capita during industrialization. These relationships survive controls for geography, literacy, schooling, and initial population and settlement characteristics. The manufacturing association is largest in 1920 and diminishes later in the twentieth century. By contrast, the association with population density remains positive through 2000, although these longer-run specifications use a less extensive set of settlement controls.

Because authors may have located in counties with favorable economic prospects, I also exploit the composition of nineteenth-century migration. Many counties, particularly in the Midwest, received substantial numbers of settlers before the main period of industrial expansion. I construct a shift-share-type instrument combining counties' birthplace shares with the prevalence of authorship among people born in each origin. The instrument predicts which counties attracted residents from places where intellectual production was more prevalent. The resulting IV estimates are positive for manufacturing specialization and subsequent invention. This approach provides evidence from the origins of county populations, alongside the comparisons conditioning on the richer set of socioeconomic controls. Its interpretation requires that the relevant settlement differences did not affect industrial outcomes through other characteristics of migrants or their chosen destinations.

The content of written works offers a further insight. Both academic and non-academic copyrighting are positively associated with manufacturing in 1860 and 1870. By 1920, the

non-academic coefficient remains positive and statistically significant, whereas the academic coefficient is imprecisely estimated near zero. Thus the relationship between writing and industrial development extends beyond works concerned with scientific knowledge and formal instruction. Copyrighting may capture intellectual characteristics of local populations that conventional measures of schooling and invention leave unobserved, without implying that every registered work contributed directly to industrial technology.

This paper contributes to the literature on the intellectual origins of industrialization. The closest empirical study is Squicciarini and Voigtländer (2015), who use subscriptions to the *Encyclopédie* to measure upper-tail human capital and study industrialization in France. Whereas subscriptions identify readers, the copyright records identify producers of registered writing and permit a comparison across subjects. The paper also relates to evidence on book production and economic growth (Baten and van Zanden, 2008), the economic effects of printing (Dittmar, 2011), and research on American invention (Sokoloff, 1988; Khan and Sokoloff, 1993) and the local determinants of structural transformation (Fiszbein, 2022).

The remainder of the paper is organized as follows. Section 2 introduces the historical setting, and Section 3 describes the copyright records and their linkage to the census. Section 4 documents the timing and geography of written production. Section 5 presents the empirical strategy, and Section 6 reports the results on manufacturing and invention. Section **??** discusses their economic interpretation. Section 7 concludes.

2 Historical setting

The first half of the nineteenth century was a period of rapid territorial expansion and internal migration in the United States. Many counties in the Midwest and trans-Appalachian West had relatively small settler populations in the early nineteenth century, before experiencing substantial population inflows (Porter et al., 1895; Vandenbroucke, 2008b). This settlement preceded much of the industrial expansion of the late nineteenth century (Gallman, 2000; Davis, 2004). In recently settled counties, the composition of the population was therefore shaped to a considerable extent by the origins of incoming migrants, rather than by an established local industrial economy.

Migration was nevertheless not random. Geographic conditions, access to transportation, and land policy influenced settlement decisions (Atack et al., 2000; Vandenbroucke, 2008a). Conditional on these factors, however, differences in the origins of settlers provide a potential source of variation in the human capital and intellectual characteristics of local populations. Migrant networks helped determine where people from particular origins settled (Bartel, 1989; Munshi, 2003), while the characteristics they brought with them could shape the development of their destinations. This distinction is particularly relevant in a period when schooling was largely organized and financed by local communities (Goldin, 2016).

Authors participated extensively in these population movements. As documented below, individuals identified in the copyright records were considerably more likely than other residents to live outside their state of birth, particularly in the Midwest. The geographic distribution of intellectual production could therefore reflect not only the economic opportunities available in

different counties, but also the intellectual characteristics of the populations that settled them. This historical setting motivates the migration-based empirical strategy developed in Section 5.

3 Measuring written production

3.1 Copyright records and author identification

I obtain copyright registrations from the Library of Congress collection of *Early Copyright Title Pages*, which covers registrations made between 1790 and 1870 (Library of Congress, n.d.). During this period, applicants submitted title pages to federal district courts and the surviving records were transferred to the Library of Congress when registration was centralized in 1870 (Khan, 2005). The title pages record the titles, dates, author names, and publication state of works submitted for copyright protection. Unlike patent records, they cover writing on a wide range of subjects, including instruction, religion, literature, and professional and technical matters. They offer a direct record of a portion of the intellectual work produced in the early United States.

I collect the transcribed title pages and extract recorded dates and author names from their text. The collection comprises approximately 50,000 works represented by more than 90,000 images, which include accompanying documents and blank pages. Dates can be identified for approximately 70,000 entries.

Names are extracted from the text following the first occurrence of "by" on a title page and compared with census names. When several authors are named, the procedure retains the first author. The county measure thus locates identified authors by residence, rather than assigning works to their place of publication.

3.2 Linking authors to the 1850 census

I link authors of works registered between 1840 and 1860 to the 1850 census, which provides individual information on residence, birthplace, and occupation for the free population. The census permits authors to be located at the county level and provides the state of birth information used in the migration-based instrument.

The matching procedure begins with the author's first and last names and, when available, the state of registration. To accommodate spelling variation, names are encoded using Soundex. The candidate pool is restricted to the professional, teaching, clerical, and technical occupations listed in Appendix Table A3, with an additional restriction that candidates must have been at least 25 years old when the work was registered. A second search across states outside the state of registration is carried out for names not matched in the initial search.

Of the 9,662 entries which occur between 1840 and 1860, 71.7% have a single eligible census candidate under these restrictions (Appendix Table A2) and are thus uniquely matched. The linked sample contains 5,162 identified authors whose characteristics are reported in Appendix Table A4. All registrations attributed to an author are assigned to that person's county of residence in 1850. This procedure locates the population associated with written production around mid-century; it does not establish where each work was composed or where the author lived at the time of every registration. Since populous counties also have more people who might

match an extracted name, the fully specified OLS regressions control for county population, in addition to the other initial characteristics described below.[1]

3.3 The subject matter of written production

The title pages also allow me to distinguish academic from non-academic writing. I define academic works as those primarily intended to teach or advance non-religious knowledge, including textbooks, instructional materials, and scientific publications. All remaining works, including novels, poetry, religious writings, biographies, political pamphlets, and other literary and popular publications, are classified as non-academic.

I classify works into these two categories using a two-stage procedure. First, I select a small set of keywords associated with academic writing, such as "textbook", "grammar", "university", and "teacher", and rank the title pages according to the relative frequency of these words. I then label the top 10% of works as academic and the bottom 10% as non-academic. These observations provide a training sample covering 20% of the corpus.

In the second stage, I estimate a logistic regression using the frequency of every word appearing in the training sample. I use the estimated model to classify the remaining 80% of works. The procedure thus uses a small set of indicative words to construct the training sample, while allowing the classification itself to draw on the full vocabulary of the title pages. Appendix D reports the keywords and classification method. I also construct narrower subject categories using separate keyword lists, allowing works to belong to more than one category.

3.4 County variables and other sources

The main county measure aggregates registrations in 1840–1860 attributed to individuals residing in the county in 1850. Authors with several registrations contribute several records to the county total. Let C_i denote this count. The regressor is $k_i = \log(1 + C_i)$, which retains counties with no linked registrations. The migration-based instrument instead uses the prevalence of authorship among people born in different origins, as described in Section 5.2.

The main outcomes are manufacturing's share of employment at selected dates and patents per capita over 1860–1940. I also examine the evolution of population density from 1860 to 2000, changes in urbanization, and income per capita in 2000. Historical economic outcomes and geographic variables come primarily from Fiszbein (2022); the census-based series on education, population, and religion come from Haines and ICPSR (2010). National patent totals are obtained from U.S. Patent and Trademark Office (2022).

County controls are measured in 1850, apart from geographic characteristics. They include literacy and school enrollment, population and density, urbanization, and the shares of foreign-born and interstate migrants. Geographic controls include climate, agricultural suitability, terrain ruggedness, and proximity to oceans or the Great Lakes. The county outcome regressions include state fixed effects and a polynomial in latitude and longitude; Appendix Table B1 provides the complete variable list.

[1] In an exploratory placebo exercise, an indicator constructed by treating 4,000 randomly selected census individuals as authors was generally unrelated to the outcomes after controlling for population.

4 Descriptive Evidence

Figure 1 compares copyright registrations and patenting relative to population between 1790 and 1870. Copyright activity begins a sustained increase around 1830, preceding the sharper rise in patenting in subsequent decades. This expansion took place as manufacturing began to grow but before the much larger industrial transformation of the late nineteenth century (Davis, 2004; Sokoloff, 1988). The evidence places the expansion of written production well before the period conventionally associated with knowledge-based economic growth (Abramovitz and David, 2000). Appendix Figure A1 reports the underlying annual series.

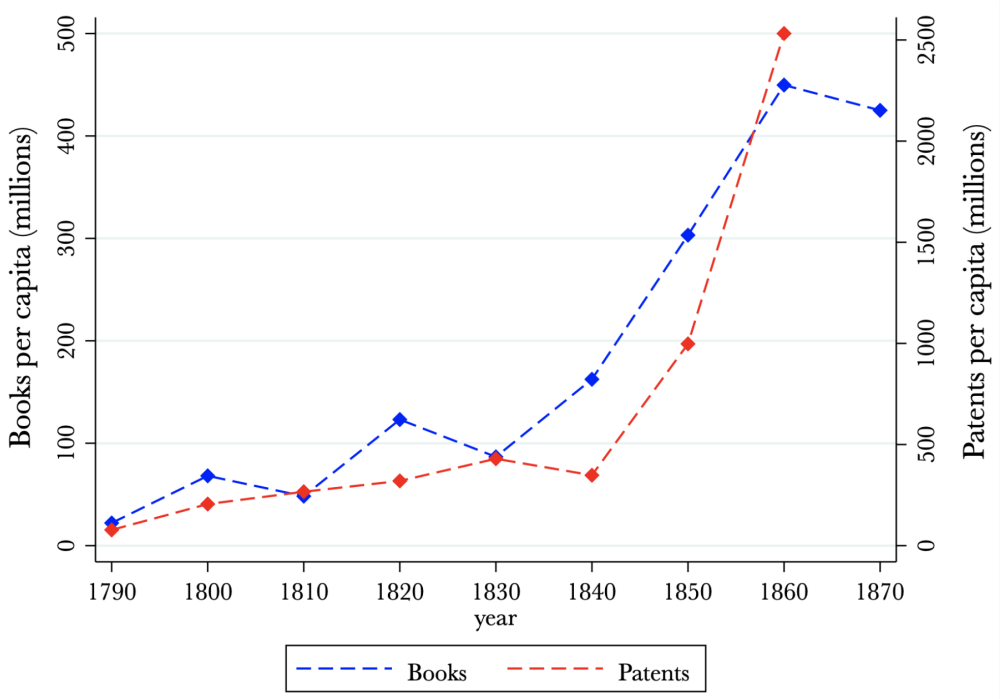


Figure 1. Copyrighting and patenting relative to population

Notes: Decadal population-normalized series. Copyrights use the left axis and patents the right axis; the axes have different scales. "Books" denotes dated copyright records. Sources: author's calculations from Library of Congress (n.d.) and national patent statistics from U.S. Patent and Trademark Office (2022).

Copyrighting was geographically concentrated, but not confined to a few large publishing centers. Figure 2 locates registered works using the residences of their authors in the 1850 census. The Northeast accounts for much of the activity, yet considerable variation appears within states and across counties farther west and in the south. The map also illustrates the geographic distribution of academic and non-academic registrations.

The distribution of copyrighting reflects part of the initial development of these counties. Places with more registrations tended to be larger, more urban, and more literate, and to have more schools and colleges (Appendix Table A1). These differences make it essential to distinguish authorship from the economic and educational characteristics already present in 1850. The next section develops this comparison and exploits variation in the birthplaces of county residents to predict local intellectual production.

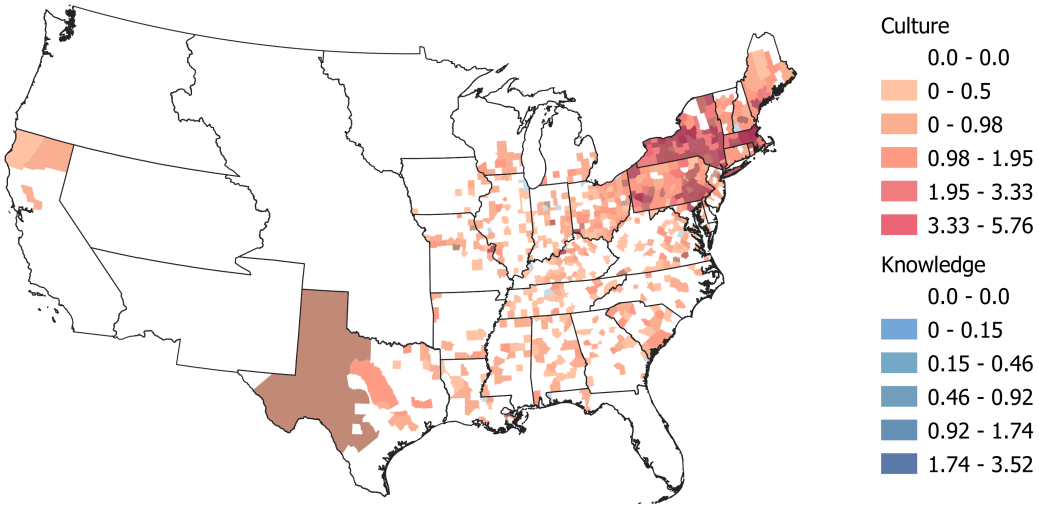


Figure 2. The county geography of linked copyright activity, 1840–1860

Notes: Registrations are assigned to the matched author's county of residence in the 1850 census. The map uses "Knowledge" (blue) for works classified as academic and "Culture" (red) for non-academic works; overlapping colors appear purple. Shading displays log-count categories. Sources: author's calculations from the copyright records and census links.

4.1 The composition of knowledge over time

Table 1 shows the evolution of the composition of knowledge production over time. Topic classification is done using the hybrid keyword and logit classification described in Section 3.3. Using a similar procedure, I breakdown the corpus into finer categories distinguishing purpose rather than subject. *Practical knowledge* contains works dedicated to the dissemination of knowledge such as textbooks and manuals for practitioners. *Theoretical knowledge* concerns the advancement of knowledge, such as in new treatises, essays and other research publications.

Table 1. Academic writing and its practical–theoretical composition

	All entries (%)		Within academic writing (%)	
Decade	Academic	Non-academic	Practical knowledge	Theoretical knowledge
1790s	7.4	92.6	77.3	22.7
1800s	6.2	93.8	100.0	0.0
1810s	12.3	87.7	74.1	25.9
1820s	18.3	81.7	75.0	25.0
1830s	21.8	78.2	78.2	21.8
1840s	22.1	77.9	80.7	19.3
1850s	16.7	83.3	81.9	18.1
1860s	14.3	85.7	83.0	17.0

Notes: The sample contains 41,715 screened title entries, including repeated editions. Source: Library of Congress.

The estimates reveal an early expansion of academic writing followed by a continuing shift towards practical application within it. The academic share rises from 12.3% in the 1810s to 22.1% in the 1840s, before declining to 14.3% in the 1860s. Meanwhile, the practical share within academic writing increases from 74.1% to 83.0% between the 1810s and the 1860s, indicating an overwhelming, and growing, emphasis on teaching readers how to apply knowledge. This combination is consistent with Mokyr's account of the economic importance of "useful knowledge,"

which emphasizes both the expansion of knowledge and its dissemination to those able to put it into practice as key drivers of economic growth (Mokyr, 2005).

5 Empirical strategy

5.1 County comparisons

The empirical analysis relates intellectual production in 1840–1860 to manufacturing specialization and inventive activity during the subsequent industrial expansion. For county i in state $s(i)$ and outcome year t, I estimate

$$y_{it} = \alpha_{s(i),t} + \beta_t k_i + X_i'\gamma_t + P_t(\ell_i, a_i) + \varepsilon_{it}, \tag{1}$$

where $k_i = \log(1 + C_i)$ measures county copyright activity, X_i contains the baseline county characteristics, and $P_t(\ell_i, a_i)$ is a third-degree polynomial in longitude and latitude. Each outcome year is estimated separately. State fixed effects restrict the comparison to counties within the same state, while geography, literacy, schooling, and settlement characteristics account for differences observable at baseline in 1850. Standard errors use the spatial correction of Conley (1999), with a 100-kilometer cutoff.

I estimate the manufacturing relationship at different dates to establish whether its importance varies with the industrial transition. Patenting is instead measured as an aggregate flow over the 1860-1940 window encompassing the Second Industrial Revolution in the United States. The OLS estimates with the complete control set allow me to assess whether early authorship predicts these outcomes beyond literacy, schooling, population, and initial urbanization. Since authors may have been attracted to places with favorable industrial prospects, I also consider variation arising from internal migration as described below.

5.2 Migration and predicted authorship

Interstate migration was extensive during the first half of the nineteenth century and particularly important for the location of authors. Among native-born residents in the 1850 census, 36% of identified authors lived outside their birth state, compared with 24% of non-authors. In the Midwest, the corresponding shares were 78% and 45% (Table 2). These differences suggest that the distribution of authors was closely connected to the composition of settlement. Consistent with evidence on migrant networks, people from a given origin were more likely to settle in destinations where others from that origin resided (Bartel, 1989; Munshi, 2003).

I construct a shift-share-type instrument combining differences in authorship rates across birthplaces with the composition of each county's residents. Let A_l be the number of linked authors born in origin l and N_l the number of census residents born there. I define the authorship rate for origin l and the national rate as

$$q_l = \frac{A_l}{N_l}, \qquad q_{US} = \frac{A_{US}}{N_{US}}, \tag{2}$$

Table 2. Interstate migration among native-born residents in 1850

	Share living outside state of birth		
Region	Non-authors	Authors	Observations
Midwest	0.45	0.78	4,584,383
Northeast	0.12	0.30	7,207,901
South	0.22	0.48	5,434,905
Total	0.24	0.36	17,227,189

Notes: An interstate migrant is a native-born individual residing in a different state from the state of birth in 1850. Observations give the size of the regional sample. Sources: 1850 Census of Population and Library of Congress copyright records.

and let π_{il} denote the fraction of county i's population born in origin l. The instrument is

$$Z_i = \sum_{l \in \mathcal{L},\, l \neq s(i)} (q_l - q_{US})\pi_{il}, \tag{3}$$

where $\mathcal{L}$ includes the U.S. birth states and the United Kingdom, France, and Germany. The foreign-origin rates are computed among U.S. residents born in those countries. Residents born in the destination state are excluded from the sum.

The instrument predicts more local authorship in counties receiving a higher share of settlers from origins where writing was relatively prevalent. The first stage is

$$k_i = \eta_{s(i)} + \rho Z_i + X_i'\delta + P(\ell_i, a_i) + v_i. \tag{4}$$

Authorship rates count individuals, whereas k_i is the log of one plus their county-level registered output. The first stage therefore tests whether counties connected to high-authorship origins also produced more copyrighted works.

5.3 Identification and specification choice

The identifying assumption is that, conditional on the included controls, settlement from high-authorship origins did not independently affect later industrial outcomes through other characteristics of migrants or the places they selected. In other words, conditional on geographic characteristics individuals from high-copyrighting states did not select locations which were systematically more conducive to industrialization. To assess this possibility, Table C1 regresses the migration-based authorship instrument on destination characteristics, progressively adding state fixed effects and a latitude–longitude polynomial.

The clearest pattern is an association between predicted authorship and *more* rugged terrain, which persists after accounting for state effects and spatial polynomials. Because industrialization generally required both flat surfaces for the establishment of large factories and, in the earliest phases, elevation changes to harness water power, it is unclear whether all else equal higher terrain ruggedness should be more or less conducive to industrial activity or economic development more broadly. Higher instrument values are also associated with proximity to the ocean or Great Lakes in the first two columns, but reassuringly this relationship becomes insignificant once the spatial polynomial is included. The negative association with temperature likewise disappears after adding state effects. Hence, the data does not indicate a systematic relationship between the origin

Table 3. Geographic selection and the authorship instrument

	(1)	(2)	(3)
Terrain ruggedness	0.811***	0.848**	0.581**
	(3.79)	(3.18)	(3.19)
Distance to ocean or Great Lakes (km)	−0.0367***	−0.0333***	−0.0102
	(−9.03)	(−5.55)	(−1.43)
Temperature	−1.069**	0.947	1.514
	(−3.22)	(1.71)	(1.58)
Annual rainfall	0.000104	−0.00642	0.00652
	(0.02)	(−0.81)	(1.24)
Land suitability	−1.744	4.738	4.760
	(−0.50)	(1.73)	(1.89)
Counties	1,507	1,507	1,507
State fixed effects	No	Yes	Yes
Latitude–longitude polynomial	No	No	Yes

Notes: The dependent variable is the migration-based authorship instrument. All columns include the five geographic characteristics shown. Parentheses report t statistics based on Conley standard errors with a 100 km cutoff. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

of migrants and selection of locations more conducive to economic development. Nevertheless, I include geographic controls, state effects, and the latitude– longitude polynomial, followed by schooling, literacy, and initial urbanization controls in the most conservative specifications.

The resulting first stages are informative. The Kleibergen–Paap statistics are 13 in the main post-1850 manufacturing specifications and 14 for patenting; the manufacturing regression for 1850 yields 23. Adding population, population density, and the shares of interstate and foreign-born migrants reduces the first-stage statistic to approximately one in the patenting, urbanization, and income regressions. These additional variables absorb much of the settlement-composition variation used by the instrument. Accordingly, I report IV estimates controlling for geography, human capital, and initial urbanization, alongside OLS estimates that include the full set of initial population and migration controls.

5.4 Interpreting the IV estimates

Because the copyright measure contains measurement error arising, for instance, from some knowledge production being absent from the archive, mistakes in extracting names or linking authors, and incompleteness of the archives, we would in general expect the IV estimates to be larger in absolute value compared to their OLS counterpart. However, because the IV estimates are based on a different set of controls and on variation in settlement composition, their larger magnitudes cannot be attributed to measurement error alone. Appendix G provides a simple illustration of how incomplete measurement may generate larger IV estimates under relatively conservative conditions. In any case, the empirical results below emphasize the direction, timing, and robustness of the copyrighting relationships rather than interpreting the IV–OLS gap as a precise measure of attenuation.

6 Intellectual production and the industrial economy

6.1 Manufacturing specialization

Counties with more copyright activity in 1840–1860 subsequently developed larger manufacturing employment shares during industrialization. Table 4 reports the OLS and IV estimates for selected years; Figure 3 shows their evolution. In the fully controlled OLS specification, the copyright coefficient is positive and statistically significant in 1850, 1870, and 1920. It reaches 0.0111 ($t = 3.50$) in 1920, implying that a one-unit difference in $\log(1 + C_i)$ is associated with a 1.11 percentage point difference in manufacturing's employment share. This relationship survives controls for geography, literacy, schooling, and the initial population and settlement characteristics of each county.

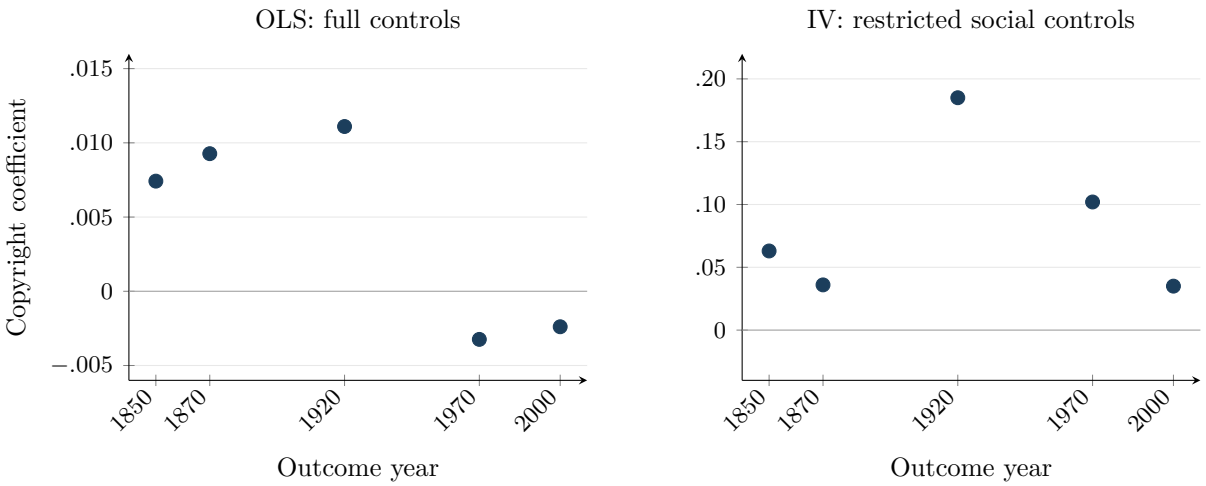


Figure 3. Early copyrighting predicts manufacturing most strongly around 1920

Notes: Selected-year point estimates from Table 4, with separate vertical scales for OLS and IV. Manufacturing employment shares are measured as fractions. Both panels include geography, human capital, baseline urbanization, state effects, and the spatial polynomial. OLS additionally includes the remaining social controls. The 1850 outcome is contemporaneous with the registration window. Sample sizes, t statistics, and first-stage diagnostics are reported in the table. No test of differences between years is reported.

The timing of this association is particularly informative. Manufacturing was already more prevalent in counties with greater written production in 1850, but the largest OLS point estimate occurs in 1920, during the period of large-scale American industrial expansion. The coefficient subsequently falls and is statistically indistinguishable from zero in 1970 and 2000. Early copyrighting thus predicts a structural transformation process with a higher manufacturing peak, rather than a persistent specialization in manufacturing activity which has been found for other drivers of development such as agricultural diversity Fiszbein (2022).

The IV estimates give a similar pattern for the component of county copyrighting predicted by settlers' origins. Their largest point estimate also occurs in 1920, when the coefficient is 0.185 ($t = 3.88$), with a Kleibergen–Paap first-stage statistic of 13. The IV coefficients decline thereafter. Their magnitude is substantially greater than the corresponding OLS estimates, although the IV specifications do not include the additional population and migration controls in the fully controlled OLS regressions as naturally these migration variables would absorb much of the variation used by the instrument.

Table 4. Copyrighting and the manufacturing employment share

Outcome year	1850	1870	1920	1970	2000
Panel A. Ordinary least squares: full controls					
$\log(1 + C_i)$	0.00742	0.00927	0.0111	−0.00324	−0.00239
	(5.65)	(3.57)	(3.50)	(−1.06)	(−0.51)
Counties	1,114	1,504	1,504	1,504	1,504
R^2	0.271	0.193	0.204	0.221	0.129
Remaining social controls	Yes	Yes	Yes	Yes	Yes
Panel B. Instrumental variables: restricted social controls					
$\log(1 + C_i)$	0.0630	0.0360	0.185	0.102	0.0350
	(16.47)	(0.44)	(3.88)	(1.96)	(0.81)
Counties	1,114	1,505	1,505	1,505	1,505
Kleibergen–Paap F statistic	23	13	13	13	13
Remaining social controls	No	No	No	No	No
State fixed effects	Yes	Yes	Yes	Yes	Yes
Latitude–longitude polynomial	Yes	Yes	Yes	Yes	Yes
Geography controls	Yes	Yes	Yes	Yes	Yes
Human capital controls	Yes	Yes	Yes	Yes	Yes
Urbanization, 1850	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is manufacturing's share of employment in the year shown. Copyright registrations refer to 1840–1860 and are assigned to authors' counties of residence in 1850. Both panels include baseline urbanization, human capital and geography controls, state fixed effects, and the spatial polynomial. The OLS panel additionally includes population, population density, and the foreign-born and internal-migrant shares; these additional controls are excluded from the IV panel. The two panels therefore differ in controls and, after 1850, slightly in sample size. Only the copyright coefficient is displayed. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff.

6.2 Subsequent invention

The geography of early copyrighting also anticipates the geography of technological invention. Table 5 relates registrations in 1840–1860 to patents per capita over 1860–1940. The fully controlled OLS coefficient is 0.195 ($t = 3.40$), remaining positive and significant after accounting for schooling, literacy, geography, urbanization, population, and migration. This is an important complement to the manufacturing result: places with more early written production subsequently generated more patents relative to their populations, not simply more manufacturing jobs.

The relationship remains positive when copyrighting is instrumented with settlers' birthplaces. The IV coefficient in the specification with geography, human capital, and baseline urbanization is 2.353 ($t = 3.59$), with a first-stage statistic of 14. The fully controlled OLS results and the settlement-based IV estimates point in the same direction. Early intellectual production predicts both manufacturing specialization and subsequent invention, two distinct features of an industrial economy.

6.3 Academic and non-academic writing

Which kinds of written production account for this relationship? The title pages permit an instructive distinction between academic and non-academic works. As described in Section 3.3, academic writing includes scientific publications and instructional materials, while the non-academic category covers the remaining subjects. Table 6 includes the two categories jointly in the fully controlled manufacturing regressions.

Table 5. Early copyrighting and subsequent invention: selected specifications

	OLS (1)	OLS (2)	IV (3)
$\log(1 + C_i)$	0.360	0.195	2.353
	(7.47)	(3.40)	(3.59)
Counties	1,499	1,498	1,499
R^2	0.181	0.223	—
Kleibergen–Paap F statistic	—	—	14
State effects and spatial polynomial	Yes	Yes	Yes
Geography and human capital	Yes	Yes	Yes
Urbanization, 1850	Yes	Yes	Yes
Remaining social controls	No	Yes	No

Notes: The outcome is log patents per capita per decade over 1860–1940. Copyright registrations cover 1840–1860 and are assigned to authors' 1850 residential counties. Columns (1) and (3) reproduce column (4) of the OLS and IV panels in Appendix Table C2; column (2) reproduces OLS column (5). The remaining social controls are log population, population density, and foreign-born and internal-migrant shares. Human-capital and social controls are measured in 1850. Parentheses report t statistics using Conley standard errors with a 100 km cutoff. The full sequence, including the weak first-stage specification, is preserved in the appendix.

Both academic and non-academic registrations are positively associated with manufacturing in 1860 and 1870 with academic registrations having a slightly larger point estimte. By 1920, the coefficient on academic writing is close to zero and imprecisely estimated, whereas the coefficient on non-academic writing remains positive at 0.0105 ($t = 3.25$). This finding extends the relationship beyond the subset of copyrighted works most closely associated with science and formal instruction. The intellectual attributes of counties relevant to industrialization were not necessarily confined to individuals writing directly about productive techniques.

Table 6. Academic and non-academic copyrighting and manufacturing employment

	(1)	(2)	(3)	(4)	(5)
Outcome year	1860	1870	1920	1970	2000
Log academic registrations	0.00899	0.00785	−0.00307	−0.00264	0.00708
	(3.34)	(2.10)	(−0.43)	(−0.45)	(1.65)
Log non-academic registrations	0.00670	0.00873	0.0105	−0.00254	−0.00246
	(4.97)	(3.50)	(3.25)	(−0.80)	(−0.54)
Observations	1504	1504	1504	1504	1504
R^2	0.295	0.195	0.203	0.221	0.130
State fixed effects	Yes	Yes	Yes	Yes	Yes
Latitude–longitude polynomial	Yes	Yes	Yes	Yes	Yes
Geographic controls	Yes	Yes	Yes	Yes	Yes
Human-capital controls	Yes	Yes	Yes	Yes	Yes
Social controls	Yes	Yes	Yes	Yes	Yes

Notes: OLS estimates. The dependent variable is the manufacturing share of employment in the indicated year. The regressors distinguish academic and non-academic registrations over 1840–1860, using the title-page classification in Appendix D. Both categories enter jointly. All columns include the full controls in Appendix Table B1, measured in 1850 where applicable. The first outcome is 1860, unlike the 1850 benchmark in Table 4. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff.

These patterns are consistent with copyrighting capturing a broader form of local intellectual activity than the production of technical knowledge. Writing on non-academic subjects may reflect a community's skills, interests, and demand for reading as well as the productive abilities of individual authors. The importance of this result is therefore not that all copyrighted works

contributed directly to manufacturing technology, but that the relationship between intellectual production and industrial development extends beyond formal academic subjects.

6.4 Population concentration beyond industrialization

The decline in the manufacturing relationship after 1920 does not imply that the initial geography of intellectual production ceased to matter. As shown in Figure 4, early copyrighting remains positively associated with county population density through 2000. The OLS relationship is relatively stable, while the IV estimates are generally larger in the later period.

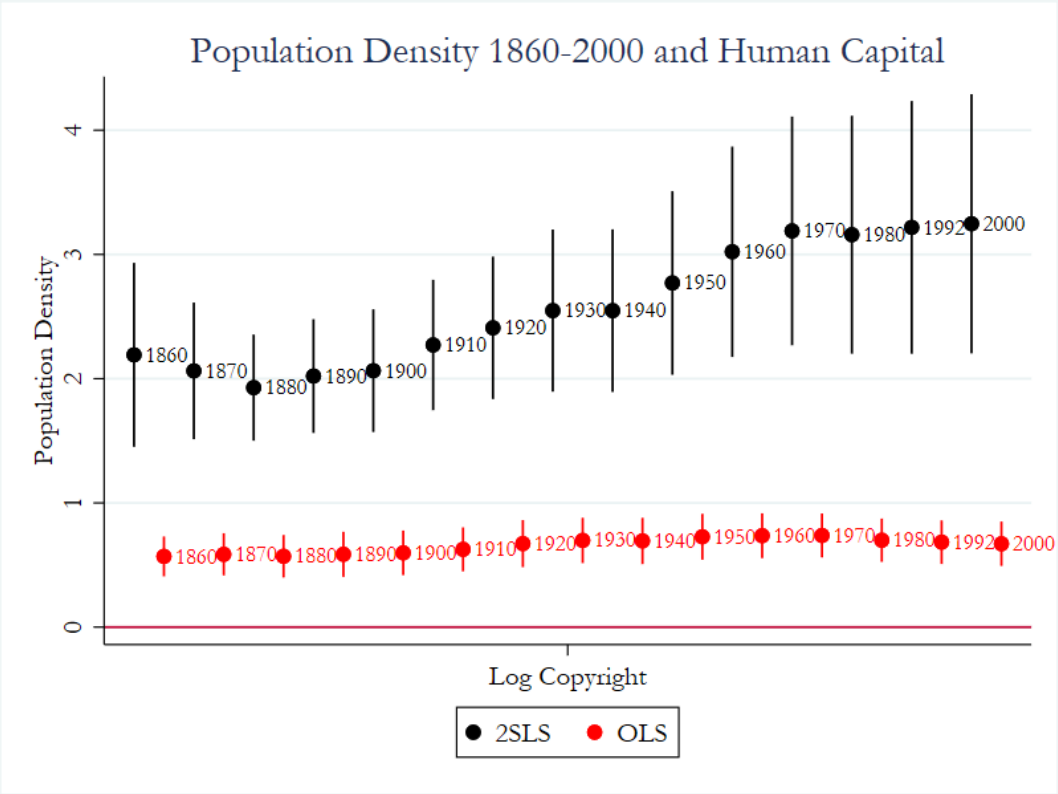


Figure 4. Copyrighting and population density, 1860–2000

Notes: Copyright coefficients and 95% confidence intervals from separate county regressions for the years shown. Black points are 2SLS and red points are OLS. The time-pattern specifications retain geography, human capital, baseline urbanization, state effects, and the spatial polynomial and exclude the remaining social controls.

Together, the manufacturing and population-density results show that early centers of intellectual production were more likely to become manufacturing centers during industrialization, and their relative population concentration persisted after manufacturing's employment share had declined. Appendix E extends the long-run analysis to urbanization showing that early copyrighting is also a strong predictor of long-run income differences across counties.

7 Conclusion

Copyright registrations per capita began a sustained rise around 1830, well before the major industrial expansion of the late nineteenth century. By linking authors active in 1840–1860 to their residences in the 1850 census, I document a geography of written production that

anticipated subsequent manufacturing specialization and inventive activity. The manufacturing relationship is largest around 1920 and fades later in the twentieth century, while the association with population density persists. A migration-based instrument using settlers' birthplaces provides complementary evidence pointing in the same direction.

The link between early writing and industrialization is not confined to scientific or academic subjects. Non-academic copyrighting also predicts manufacturing specialization, suggesting that the intellectual environment conducive to industrial development included more than the production of directly applicable technical knowledge. Copyright records offer a new way to study where intellectual activity took place before industrialization and how that geography related to the subsequent development of American industry.

References

Abramitzky, R., Boustan, L. P., and Eriksson, K. (2012). Europe's tired, poor, huddled masses: Self-selection and economic outcomes in the Age of Mass Migration. *American Economic Review*, 102(5), 1832–1856.

Abramitzky, R., Boustan, L. P., and Eriksson, K. (2014). A nation of immigrants: Assimilation and economic outcomes in the Age of Mass Migration. *Journal of Political Economy*, 122(3), 467–506.

Abramovitz, M., and David, P. A. (2000). American macroeconomic growth in the era of knowledge-based progress: The long-run perspective. In S. L. Engerman and R. E. Gallman (eds.), *The Cambridge Economic History of the United States*, Vol. 3, pp. 1–92. Cambridge University Press.

Acemoglu, D. (2002). Technical change, inequality, and the labor market. *Journal of Economic Literature*, 40(1), 7–72.

Acemoglu, D. (2003). Factor prices and technical change: From induced innovations to recent debates. In *Knowledge, Information and Expectations in Modern Macroeconomics: In Honor of Edmund S. Phelps*. Princeton University Press.

Acemoglu, D., Johnson, S., and Robinson, J. A. (2001). The colonial origins of comparative development: An empirical investigation. *American Economic Review*, 91(5), 1369–1401.

Ager, P., and Brückner, M. (2013). Cultural diversity and economic growth: Evidence from the US during the Age of Mass Migration. *European Economic Review*, 64, 76–97.

Alesina, A., and La Ferrara, E. (2005). Ethnic diversity and economic performance. *Journal of Economic Literature*, 43(3), 762–800.

Allen, R. C. (2009). *The British Industrial Revolution in global perspective*. Cambridge University Press.

Arellano, M., and Bover, O. (1995). Another look at the instrumental variable estimation of error-components models. *Journal of Econometrics*, 68(1), 29–51.

Atack, J., Bateman, F., and Parker, W. (2000). Northern agriculture and the westward movement. In S. L. Engerman and R. E. Gallman (eds.), *The Cambridge Economic History of the United States*, Vol. 2, pp. 285–328. Cambridge University Press.

Bartel, A. P. (1989). Where do the new U.S. immigrants live? *Journal of Labor Economics*, 7(4), 371–391.

Baten, J., and van Zanden, J. L. (2008). Book production and the onset of modern economic growth. *Journal of Economic Growth*, 13, 217–235.

Bazzi, S., Ferrara, A., Fiszbein, M., Pearson, T., and Testa, P. A. (2022). Sundown towns and racial exclusion: The Southern white diaspora and the "Great Retreat". *AEA Papers and Proceedings*, 112, 234–238.

Bazzi, S., Ferrara, A., Fiszbein, M., Pearson, T., and Testa, P. A. (2023). The other great migration: Southern whites and the New Right. *Quarterly Journal of Economics*, 138(3), 1577–1647.

Bazzi, S., Fiszbein, M., and Gebresilasse, M. (2020). Frontier culture: The roots and persistence of 'rugged individualism' in the United States. *Econometrica*, 88(6), 2329–2368.

Bazzi, S., Fiszbein, M., and Gebresilasse, M. (2021). Rugged individualism and collective (in)action during the COVID-19 pandemic. *Journal of Public Economics*, 195, 104357.

Becker, S. O., and Woessmann, L. (2009). Was Weber wrong? A human capital theory of Protestant economic history. *Quarterly Journal of Economics*, 124(2), 531–596.

Bisin, A., and Verdier, T. (2001). The economics of cultural transmission and the dynamics of preferences. *Journal of Economic Theory*, 97(2), 298–319.

Blundell, R., and Bond, S. (1998). Initial conditions and moment restrictions in dynamic panel data models. *Journal of Econometrics*, 87(1), 115–143.

Cantoni, D. (2015). The economic effects of the Protestant Reformation: Testing the Weber hypothesis in the German lands. *Journal of the European Economic Association*, 13(4), 561–598.

Cantoni, D., Dittmar, J., and Yuchtman, N. (2018). Religious competition and reallocation: The political economy of secularization in the Protestant Reformation. *Quarterly Journal of Economics*, 133(4), 2037–2096.

Chaney, E. (2016). Religion and the rise and fall of Islamic science. Working paper, Harvard University, May.

Clark, G. (2008). *A Farewell to Alms: A Brief Economic History of the World*. Princeton University Press.

Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92(1), 1–45.

Cozzens, P. (2016). *The earth is weeping: The epic story of the Indian Wars for the American West*. Knopf.

Davis, J. H. (2004). An annual index of U.S. industrial production, 1790–1915. *Quarterly Journal of Economics*, 119(4), 1177–1215.

De Long, J. B., and Shleifer, A. (1993). Princes and merchants: European city growth before the Industrial Revolution. *Journal of Law and Economics*, 36(2), 671–702.

Dittmar, J. E. (2011). Information technology and economic change: The impact of the printing press. *Quarterly Journal of Economics*, 126(3), 1133–1172.

Doepke, M., and Zilibotti, F. (2008). Occupational choice and the spirit of capitalism. *Quarterly Journal of Economics*, 123(2), 747–793.

Fernández, R., and Rogerson, R. (1996). Income distribution, communities, and the quality of public education. *Quarterly Journal of Economics*, 111(1), 135–164.

Fernández, R., and Rogerson, R. (1998). Public education and income distribution: A dynamic quantitative evaluation of education-finance reform. *American Economic Review*, 88(4), 813–833.

Ferrie, J. P. (1999). *Yankeys now: Immigrants in the antebellum United States, 1840–1860*. Oxford University Press.

Fiszbein, M. (2022). Agricultural diversity, structural change, and long-run development: Evidence from the United States. *American Economic Journal: Macroeconomics*, 14(2), 1–43.

Gallman, R. E. (2000). Economic growth and structural change in the long nineteenth century. In S. L. Engerman and R. E. Gallman (eds.), *The Cambridge Economic History of the United States*, Vol. 2, pp. 1–56. Cambridge University Press.

Galor, O. (2011). *Unified growth theory*. Princeton University Press.

Galor, O., and Moav, O. (2002). Natural selection and the origin of economic growth. *Quarterly Journal of Economics*, 117(4), 1133–1191.

Glaeser, E. L., La Porta, R., Lopez-de-Silanes, F., and Shleifer, A. (2004). Do institutions cause growth? *Journal of Economic Growth*, 9(3), 271–303.

Goldin, C. (2016). Human capital. In C. Diebolt and M. Haupert (eds.), *Handbook of Cliometrics*, pp. 55–86. Springer.

Haines, M. R., and Inter-university Consortium for Political and Social Research (2010). *Historical, Demographic, Economic, and Social Data: The United States, 1790–2002*. ICPSR 2896, version 3. https://doi.org/10.3886/ICPSR02896.v3.

Jelodar, H., Wang, Y., Yuan, C., Feng, X., Jiang, X., Li, Y., and Zhao, L. (2019). Latent Dirichlet allocation (LDA) and topic modeling: Models, applications, a survey. *Multimedia Tools and Applications*, 78, 15169–15211.

Jones, C. I. (2001). Was an Industrial Revolution inevitable? Economic growth over the very long run. *Advances in Macroeconomics*, 1(2), Article 1.

Jones, C. I. (2022). The past and future of economic growth: A semi-endogenous perspective. *Annual Review of Economics*, 14, 125–152.

Kelly, M. (2020). Understanding persistence. CEPR Discussion Paper 15246.

Kelly, M., Mokyr, J., and Ó Gráda, C. (2014). Precocious Albion: A new interpretation of the British Industrial Revolution. *Annual Review of Economics*, 6, 363–389.

Khan, B. Z. (2005). *The Democratization of Invention: Patents and Copyrights in American Economic Development, 1790–1920*. Cambridge University Press.

Khan, B. Z., and Sokoloff, K. L. (1993). "Schemes of practical utility": Entrepreneurship and innovation among "great inventors" in the United States, 1790–1865. *Journal of Economic History*, 53(2), 289–307.

Kremer, M. (1993). Population growth and technological change: One million B.C. to 1990. *Quarterly Journal of Economics*, 108(3), 681–716.

Library of Congress (n.d.). *Early Collection of Copyright Title Pages: 1790–1870*. https://www.loc.gov/item/2019713455/.

Mao, Y., and Wang, J. J. (2022). Access to finance and technological innovation: Evidence from pre-Civil War America. Working paper, revised May 2022. https://doi.org/10.2139/ssrn.3143199.

Mitch, D. F. (1993). The role of human capital in the first Industrial Revolution. In J. Mokyr (ed.), *The British Industrial Revolution: An Economic Perspective*, pp. 267–307. Westview Press.

Mokyr, J. (2002). *The Gifts of Athena: Historical Origins of the Knowledge Economy*. Princeton University Press.

Mokyr, J. (2005). The intellectual origins of modern economic growth. *Journal of Economic History*, 65(2), 285–351.

Mokyr, J. (2009). *The Enlightened Economy: An Economic History of Britain, 1700–1850*. Yale University Press.

Mokyr, J. (2016). *A Culture of Growth*. Princeton University Press.

Montalvo, J. G., and Reynal-Querol, M. (2003). Religious polarization and economic development. *Economics Letters*, 80(2), 201–210.

Montalvo, J. G., and Reynal-Querol, M. (2005). Ethnic polarization, potential conflict, and civil wars. *American Economic Review*, 95(3), 796–816.

Munshi, K. (2003). Networks in the modern economy: Mexican migrants in the U.S. labor market. *Quarterly Journal of Economics*, 118(2), 549–599.

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2), 187–204.

Porter, R. P., Gannett, H., and Hunt, W. C. (1895). Progress of the nation, 1790 to 1890. In *Report on Population of the United States at the Eleventh Census: 1890*, Part I. Government Printing Office. https://www.census.gov/library/publications/1895/dec/volume-1.html.

Romer, P. M. (1990). Endogenous technological change. *Journal of Political Economy*, 98(5, part 2), S71–S102.

Sequeira, S., Nunn, N., and Qian, N. (2020). Immigrants and the making of America. *Review of Economic Studies*, 87(1), 382–419.

Shen, Z., Zhang, R., Dell, M., Lee, B. C. G., Carlson, J., and Li, W. (2021). LayoutParser: A unified toolkit for deep learning based document image analysis. arXiv:2103.15348. https://arxiv.org/abs/2103.15348.

Smith, J. R. (1925). *North America: Its people and the resources, development, and prospects of the continent as an agricultural, industrial, and commercial area*. Harcourt, Brace.

Sokoloff, K. L. (1988). Inventive activity in early industrial America: Evidence from patent records, 1790–1846. *Journal of Economic History*, 48(4), 813–850.

Squicciarini, M. P., and Voigtländer, N. (2015). Human capital and industrialization: Evidence from the Age of Enlightenment. *Quarterly Journal of Economics*, 130(4), 1825–1883.

U.S. Patent and Trademark Office (2022). *U.S. Patent Statistics Report*. https://www.uspto.gov/web/offices/ac/ido/oeip/taf/h_counts.htm.

Vandenbroucke, G. (2008a). The American frontier: Technology versus immigration. *Review of Economic Dynamics*, 11(2), 283–301.

Vandenbroucke, G. (2008b). The U.S. westward expansion. *International Economic Review*, 49(1), 81–110.

Voigtländer, N., and Voth, H.-J. (2006). Why England? Demographic factors, structural change and physical capital accumulation during the Industrial Revolution. *Journal of Economic Growth*, 11(4), 319–361.

A Copyright records, county measurement, and census linkage

A.1 From title pages to dated records

The Library of Congress collection covers copyright title pages from 1790–1870 (Library of Congress, n.d.). Its archival description estimates approximately 50,000 works represented in more than 90,000 images, including accompanying forms and blank pages. Processing the transcriptions yields approximately 70,000 dated entries. Dates are extracted by converting Roman numerals to Arabic numerals and identifying numerical sequences beginning with "17" or "18". The national series counts dated records rather than distinct works: renewals, resubmissions, and duplicate scans can produce multiple entries for a work.

Figure A1 reports the annual counts underlying the population-normalized series in the main text. Recorded activity rises from low initial levels, grows especially rapidly around the middle of the nineteenth century, with a decline in the early 1860s interrupting this longer-run expansion.

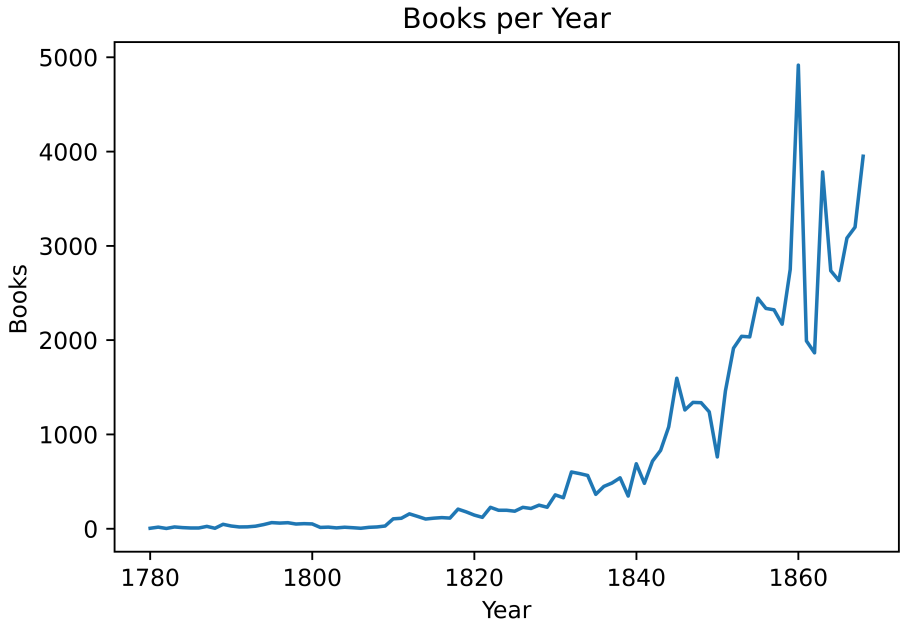


Figure A1. Annual copyright records, 1790–1870

Notes: Counts of dated records constructed from Library of Congress title-page transcriptions. Records can include renewals, resubmissions, and duplicate scans. Source: author's calculations from Library of Congress (n.d.).

A.2 County characteristics

Registrations dated 1840–1860 are assigned to a linked author's county of residence in the 1850 census. The resulting measure associates twenty years of recorded writing with a common mid-century population; it does not locate each registration at the author's residence in its publication year. Authors with multiple registrations contribute multiple records to their county's total.

Table A1 compares counties above and below the median number of registrations in the descriptive sample. Above-median counties have larger populations, higher literacy and urban-

ization, more schools per child, and more colleges. These differences motivate the controls for initial population, settlement, and education in the county comparisons.

Table A1. County characteristics above and below median copyrighting

	Above median		Below median		Difference
	Mean	SD	Mean	SD	t statistic
Population	25797.150	33419.058	8367.849	5061.875	−12.923
Literacy rate	0.876	0.114	0.792	0.151	−11.134
Urban population share	0.021	0.120	0.000	0.000	−4.284
Enslaved population share	0.167	0.223	0.211	0.226	3.433
Number of colleges	0.306	0.720	0.053	0.230	−8.390
Schools per child	0.020	0.014	0.015	0.012	−5.605
Counties	628		629		1257

Notes: Counties are divided at the median number of copyright registrations. The final column reports difference-in-means t statistics, with the below-median group first in the comparison. Literacy, urbanization, and the enslaved population share are fractions. Selected characteristics are shown. Sources: copyright records from the Library of Congress; county characteristics from Haines and ICPSR (2010).

A.3 Matching procedure and candidate multiplicity

Author names are extracted from the text following the first occurrence of "by" and compared with first and last names in the 1850 census. For works with multiple authors, the procedure identifies the first author. When the publication state can be identified, the initial census search is restricted to that state. Soundex coding accommodates spelling variation. Candidates must belong to the occupations in Table A3 and have an implied age of at least 25 when the work was produced. A second search extends to other states for names left unmatched by the initial search. The census record supplies the residential county.

Table A2 reports candidate multiplicity for 9,662 matching entries. Of these, 6,927, or 71.69%, have a single eligible census candidate. Uniqueness is conditional on the name, state, occupation, and age restrictions; it is not an independent validation of the link. The table concerns matching entries, whereas Table A4 describes 5,162 linked individuals.

Table A2. Candidate multiplicity after the census matching restrictions

Candidate matches	Frequency	Percent
1	6927	71.69
2	990	10.25
3	585	6.055
4	364	3.767
5	255	2.639
6	180	1.863
7	133	1.377
8	88	0.911
9	90	0.931
10	50	0.517
Total	9662	100

Notes: The first column reports the number of census candidates associated with a name after applying the matching restrictions. A single candidate denotes a unique match under the stated name, location, occupation, and age restrictions. Sources: Library of Congress copyright records and the 1850 Census of Population.

A.4 The eligible occupational pool and linked authors

Table A3 lists the eligible census occupations and their frequencies. The pool includes professional and technical occupations such as teachers, clergy, physicians, and lawyers. This screen limits the search to people whose reported work is consistent with authorship, and defines the non-author comparison group in Table A4. It also conditions which writers can enter the linked sample.

Table A3. Occupations eligible for author-to-census matching

Occupation	Frequency	Percent
Accountants and auditors	1215	0.802
Architects	572	0.377
Artists and art teachers	2238	1.477
Authors	96	0.0634
Chemists	436	0.288
Clergymen	27698	18.28
College presidents and deans	79	0.0521
Agricultural sciences	1	0.000660
Chemistry	21	0.0139
Engineering	2	0.00132
Geology and geophysics	2	0.00132
Mathematics	32	0.0211
Medical sciences	11	0.00726
Physics	3	0.00198
Natural science (n.e.c.)	8	0.00528
Social sciences (n.e.c.)	12	0.00792
Nonscientific subjects	160	0.106
Subject not specified	539	0.356
Dentists	3029	1.999
Designers	113	0.0746
Draftsmen	104	0.0686
Editors and reporters	1526	1.007
Engineers, civil	812	0.536
Engineers, electrical	1	0.000660
Engineers, industrial	3	0.00198
Engineers, mechanical	49	0.0323
Engineers, mining	5	0.00330
Engineers (n.e.c.)	37	0.0244
Lawyers and judges	25129	16.58
Musicians and music teachers	3808	2.513
Nurses, professional	6	0.00396
Agricultural scientists	7	0.00462
Biological scientists	55	0.0363
Geologists and geophysicists	27	0.0178
Physicists	13	0.00858
Miscellaneous natural scientists	13	0.00858
Pharmacists	6209	4.098
Physicians and surgeons	44721	29.51
Religious workers	406	0.268
Statisticians and actuaries	4	0.00264
Teachers (n.e.c.)	31217	20.60
Technicians, medical and dental	3	0.00198
Technicians, testing	23	0.0152
Technicians (n.e.c.)	3	0.00198
Therapists and healers (n.e.c.)	370	0.244
Veterinarians	142	0.0937
Professional, technical and kindred workers (n.e.c.)	371	0.245
Opticians and lens grinders and polishers	200	0.132
Total	151531	100

Notes: Census occupational categories and frequencies reproduce the set eligible for matching. "N.e.c." means not elsewhere classified. Frequencies describe the census population eligible for matching. Source: 1850 Census of Population.

Relative to other individuals in the eligible occupations, linked authors are older and report higher real estate values. Their foreign-born share is 15.8%, compared with 11.8% among eligible

non-authors. These comparisons describe differences within the occupational pool, rather than between authors and the population as a whole.

Table A4. Linked authors and other census residents in eligible occupations

	Authors		Non-authors		Difference
	Mean	SD	Mean	SD	t statistic
White	0.990	0.098	0.995	0.073	3.105
Real estate value	2395.759	11704.767	1679.085	10034.188	−4.343
Foreign-born	0.158	0.364	0.118	0.323	−7.658
Age	38.810	12.924	35.834	12.731	−16.271
Observations	5162		146369		151531

Notes: Non-authors are individuals in the eligible occupations in Table A3 who are not linked to copyright registrations. White and foreign-born are binary indicators; real estate value is measured in 1850 dollars. The final column reports difference-in-means t statistics, with non-authors first in the comparison. Source: 1850 Census of Population linked to Library of Congress copyright records.

B Variables and data sources

The copyright measures have three distinct units. National series count dated records in the 1790–1870 archive. County copyrighting counts registrations from 1840–1860 assigned to authors' 1850 residences. The regressions use $k_i = \log(1 + C_i)$, retaining counties with no linked registrations. The instrument instead uses the number of authors relative to census residents from each birthplace, as defined in equations (2)–(3).

Historical economic outcomes and geographic variables are drawn primarily from Fiszbein (2022). Census-based urbanization, education, and church counts come from Haines and ICPSR (2010); national patent totals come from U.S. Patent and Trademark Office (2022). Table B1 lists the covariate groups. Human-capital and social characteristics are measured in 1850. The nested specifications introduce baseline urbanization separately before the remaining population and migration controls.

Manufacturing is measured as its share of employment in each outcome year. The patenting outcome is the log of patents per capita per decade over 1860–1940. Urban population-share growth is the urban share in 1940 minus the urban share in 1860. The income outcome is log income per capita in 2000. The population-density estimates cover 1860–2000. Samples vary with the outcome and the availability of the included controls, as reported in each table.

Table B1. Control groups used in the county regressions

Group	Variables
Geography	Potential yields; average rainfall; temperature; distance to oceans or Great Lakes; terrain ruggedness.
Human capital	Literacy rate; school enrollment rate.
Social characteristics	Log population; urbanization; fraction of foreign-born residents; internal migrants; population density.

Notes: Human-capital and social characteristics are measured in 1850. Specifications also include state fixed effects and a third-degree polynomial in latitude and longitude. In the sequence of nested specifications, urbanization enters before the remaining social characteristics. Sources: Fiszbein (2022) and Haines and ICPSR (2010).

C Instrument balance and sensitivity to settlement controls

C.1 Geographic balance

Table C1 examines whether the migration-based prediction of authorship varies systematically with destination geography. The regressions take the form

$$Z_i = \mu_{s(i)} + G_i'\theta + P(\ell_i, a_i) + u_i, \tag{C1}$$

with state effects and the third-degree latitude–longitude polynomial introduced sequentially. The exercise describes how the settlement-based index varies with observable geographic characteristics.

Ruggedness is positively associated with the instrument in all three columns and is the only geographic characteristic significant at the 5% level after including both state effects and the spatial polynomial. Distance to the ocean or Great Lakes has a negative association in the first two columns, which becomes insignificant with the polynomial. Temperature changes sign and loses significance with state effects; rainfall and land suitability are insignificant at the 5% level throughout. The main IV specifications retain these geographic controls.

Table C1. Geographic balance of the migration-based authorship instrument

	(1)	(2)	(3)
Terrain ruggedness	0.811	0.848	0.581
	(3.79)	(3.18)	(3.19)
Distance to ocean or Great Lakes (km)	−0.0367	−0.0333	−0.0102
	(−9.03)	(−5.55)	(−1.43)
Temperature	−1.069	0.947	1.514
	(−3.22)	(1.71)	(1.58)
Annual rainfall	0.000104	−0.00642	0.00652
	(0.02)	(−0.81)	(1.24)
Land suitability	−1.744	4.738	4.760
	(−0.50)	(1.73)	(1.89)
Counties	1,507	1,507	1,507
R^2	0.423	0.145	0.235
State fixed effects	No	Yes	Yes
Latitude–longitude polynomial	No	No	Yes

Notes: The dependent variable is the migration-based instrument. Column (3) includes a third-degree polynomial in latitude and longitude. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff.

C.2 The full patenting specification sequence

Table C2 reports the complete control sequence behind the patenting results. The OLS coefficient falls from 0.486 with state effects and the spatial polynomial to 0.195 with all controls, but remains positive and statistically significant ($t = 3.40$ in the final column). Thus the patenting association survives conditioning on the observed population and migration characteristics that substantially reduce its size.

The first four IV coefficients range from 1.344 to 2.353, with Kleibergen–Paap statistics of 14–28. Adding the remaining social controls reduces that statistic to 1. The resulting coefficient, 53.89 with a t statistic of 0.10, is too imprecise to support a conclusion about the magnitude of the relationship. The urbanization and income sequences in Appendix E show the same loss of

first-stage strength. These specifications reveal how much identifying variation is absorbed by controls for initial settlement. Excluding those controls preserves first-stage strength but places more weight on the conditional exclusion restriction discussed in the main text.

Table C2. Copyrighting and patenting, 1860–1940: complete control sequence

	(1)	(2)	(3)	(4)	(5)
Panel A. Ordinary least squares					
$\log(1 + C_i)$	0.486	0.441	0.400	0.360	0.195
	(9.30)	(12.28)	(10.55)	(7.47)	(3.40)
Counties	1,506	1,501	1,499	1,499	1,498
R^2	0.104	0.147	0.179	0.181	0.223
Panel B. Instrumental variables					
$\log(1 + C_i)$	1.448	1.416	1.344	2.353	53.89
	(4.48)	(4.54)	(4.51)	(3.59)	(0.10)
Counties	1,506	1,501	1,499	1,499	1,498
Kleibergen–Paap F statistic	24	28	22	14	1
State fixed effects	Yes	Yes	Yes	Yes	Yes
Latitude–longitude polynomial	Yes	Yes	Yes	Yes	Yes
Geography controls	No	Yes	Yes	Yes	Yes
Human capital controls	No	No	Yes	Yes	Yes
Urbanization, 1850	No	No	No	Yes	Yes
Remaining social controls	No	No	No	No	Yes

Notes: The dependent variable is the log of patents per capita per decade over 1860–1940. Copyright registrations refer to 1840–1860 and are assigned to authors' counties of residence in 1850. Only the copyright coefficient is displayed. Column (5) adds the remaining social controls to the specification in column (4). Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff.

C.3 Additional spatial checks

Additional checks using alternative latitude–longitude polynomials and standard errors clustered by state give similar qualitative results.

D Classification of written activity

D.1 Academic and non-academic writing

The binary classification separates writing directed toward transmitting or advancing knowledge that is not primarily religious in content from other written production. The academic category includes instructional materials and is therefore broader than scientific research. Authorship by a member of the clergy does not by itself determine classification: a teaching manual can be classified as academic.

Title pages are first ranked by the combined relative frequency of the terms in Table D1. The top and bottom deciles provide, respectively, the positive and negative labels for a logistic classifier fitted to title-page word counts. The classifier then assigns the remaining entries. For word-count vector x_j on title page j,

$$\Pr(D_j = 1 \mid x_j) = \Lambda(\alpha + x_j' b), \qquad \Lambda(v) = \frac{\exp(v)}{1 + \exp(v)}, \tag{D1}$$

where D_j is the keyword-derived label in the training subset. This procedure extends labels initialized from the listed vocabulary; the resulting categories should be read in that operational sense.

Table D1. Terms used to initialize the academic classification

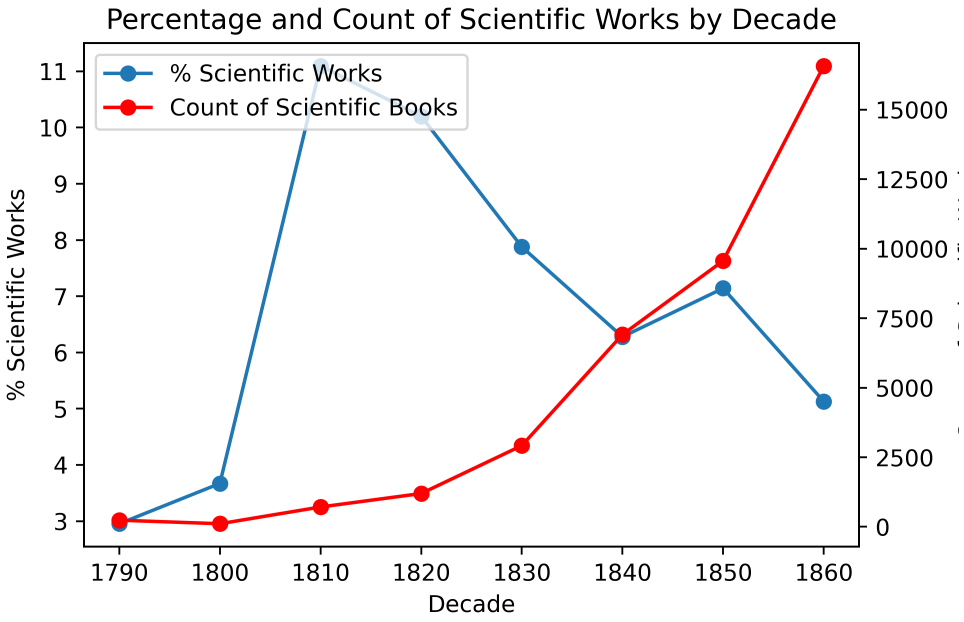
textbook	reader	instruction
principal	grammar	school
Sundayschool	Sabbath school	manual
teacher	university	professor

Notes: Terms are reproduced from the keyword list. The list is used to rank title pages before selecting the top and bottom deciles for classification.

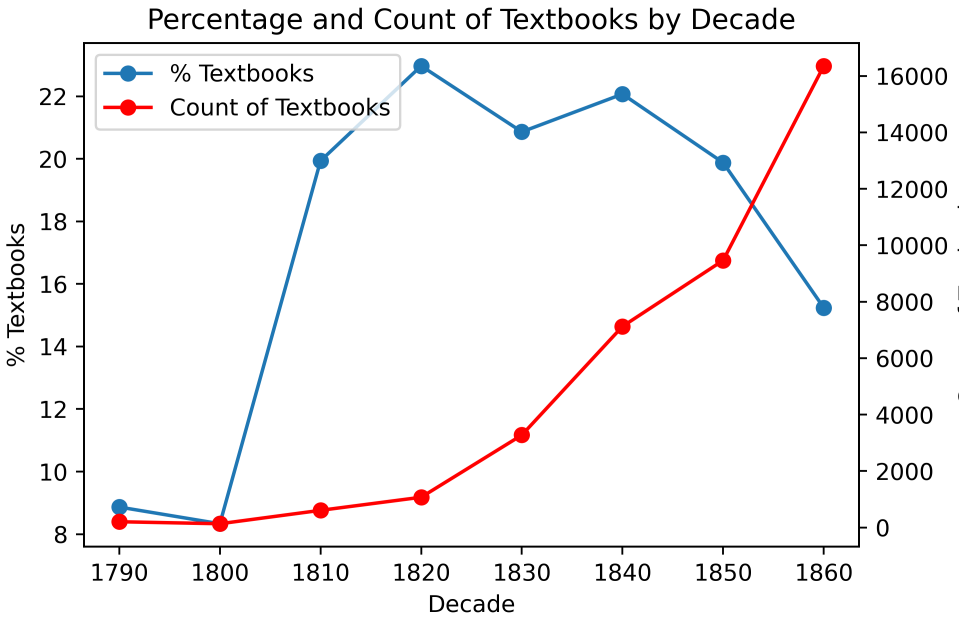
The county map and the composition regressions in Table 6 use this mutually exclusive classification. The aggregate regressions sum registrations across both categories. Separate keyword flags identify narrower topics and allow overlap between subject categories.

D.2 Supplementary subject series

Figure D1 reports counts and relative frequencies for scientific and textbook-oriented titles. These series describe both the expansion of each topic and its share of recorded production. They use subject-specific keyword flags, rather than the binary classification used in the manufacturing composition regressions.



(a) Scientific titles



(b) Textbook-oriented titles

Figure D1. Scientific and textbook-oriented writing: counts and shares

Notes: Count and relative-frequency series. The panels use different subject-specific keyword labels. The manufacturing regressions use the separate binary academic/non-academic classification. Source: author's classification of copyright title-page text.

E Supplementary development outcomes and county correlates

Population and urban growth provide broader indicators of local development, a common use of these outcomes in historical work (De Long and Shleifer, 1993). The following comparisons ask how far the copyright relationship extends beyond manufacturing and invention.

E.1 Urbanization

Table E1 relates copyrighting to the change in the urban population share from 1860 to 1940. Every column conditions on the 1850 urban share. With geography and human capital included, the OLS coefficient is 0.0548 and the IV coefficient is 0.358; both are statistically significant. The first three columns yield OLS estimates of 0.0548–0.0654 and IV estimates of 0.328–0.365.

The relationship weakens substantially with the remaining population and migration controls. The final OLS coefficient is 0.0169 ($t = 1.11$), and the IV first-stage statistic falls to 1. Initial settlement characteristics thus account for much of the conditional urbanization association. This result differs from the manufacturing and patenting relationships, which remain visible in the fully controlled OLS specifications.

Table E1. Copyrighting and the change in urban population share, 1860–1940

	(1)	(2)	(3)	(4)
Panel A. Ordinary least squares				
$\log(1 + C_i)$	0.0654	0.0617	0.0548	0.0169
	(4.45)	(4.36)	(3.92)	(1.11)
Urban population share, 1850	−0.503	−0.504	−0.482	−0.712
	(−7.98)	(−7.66)	(−7.51)	(−8.02)
Counties	1,472	1,467	1,466	1,465
Panel B. Instrumental variables				
$\log(1 + C_i)$	0.328	0.365	0.358	−2.832
	(2.38)	(2.45)	(2.13)	(−0.05)
Urban population share, 1850	−1.200	−1.279	−1.260	1.954
	(−3.44)	(−3.62)	(−3.12)	(0.04)
Counties	1,472	1,467	1,466	1,465
Kleibergen–Paap F statistic	16	17	13	1
State fixed effects	Yes	Yes	Yes	Yes
Latitude–longitude polynomial	Yes	Yes	Yes	Yes
Geography controls	No	Yes	Yes	Yes
Human capital controls	No	No	Yes	Yes
Urbanization, 1850	Yes	Yes	Yes	Yes
Remaining social controls	No	No	No	Yes

Notes: The outcome is growth in the urban population share between 1860 and 1940. Copyright registrations refer to 1840–1860 and are assigned to authors' counties of residence in 1850. All columns include baseline urbanization; the final column additionally includes the remaining social controls. Only selected coefficients are displayed. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff.

E.2 Income per capita in 2000

Table E2 extends the comparison to income per capita in 2000. Both OLS and IV estimates are positive and statistically significant through column (4), which includes geography, human capital, and baseline urbanization. With the remaining social controls, the OLS coefficient is 0.0135 ($t = 1.04$). The IV first-stage statistic falls to 1, and the corresponding estimate is highly

imprecise. As with urbanization, the income results are sensitive to the treatment of initial settlement.

Table E2. Copyrighting and income per capita in 2000

	(1)	(2)	(3)	(4)	(5)
Panel A. Ordinary least squares					
$\log(1 + C_i)$	0.0861	0.0704	0.0629	0.0500	0.0135
	(7.09)	(5.43)	(5.13)	(4.42)	(1.04)
R^2	0.0877	0.155	0.178	0.183	0.232
Panel B. Instrumental variables					
$\log(1 + C_i)$	0.417	0.421	0.406	0.782	7.988
	(6.12)	(5.82)	(5.14)	(3.24)	(0.12)
Kleibergen–Paap F	24	28	22	13	1
Observations (both panels)	1512	1507	1505	1505	1504
State fixed effects	Yes	Yes	Yes	Yes	Yes
Latitude–longitude polynomial	Yes	Yes	Yes	Yes	Yes
Geographic controls	No	Yes	Yes	Yes	Yes
Human-capital controls	No	No	Yes	Yes	Yes
Baseline urbanization	No	No	No	Yes	Yes
Remaining social controls	No	No	No	No	Yes

Notes: The dependent variable is log income per capita in 2000. Copyright registrations cover 1840–1860. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff. Column (5) adds the full social-control group, including log population, to the controls in column (4). The control groups are listed in Table B1.

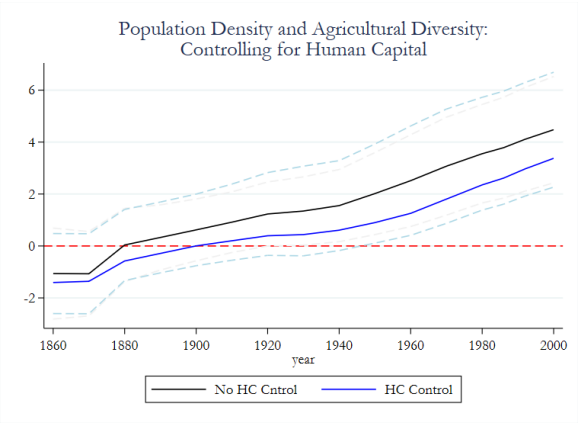
E.3 Agricultural diversity

The joint specification with agricultural diversity connects the copyright evidence to Fiszbein (2022):

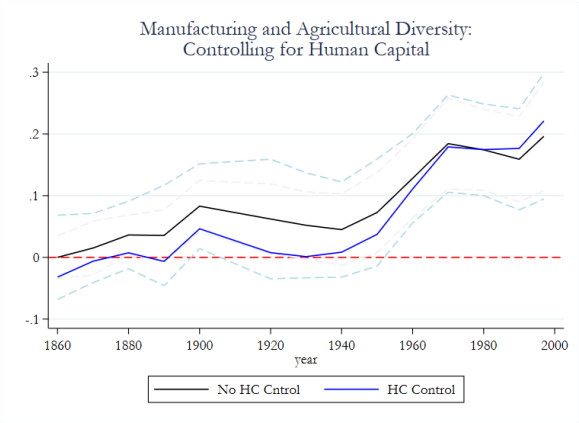
$$y_{it} = \alpha_{s(i),t} + \beta_t k_i + \gamma_t A_i + X_i' \delta_t + \varepsilon_{it}, \tag{E1}$$

where A_i is agricultural diversity in 1860. Copyrighting is instrumented with the migration-composition index, and agricultural diversity with the instrument in Fiszbein (2022). The exercise retains that analysis's baseline controls.

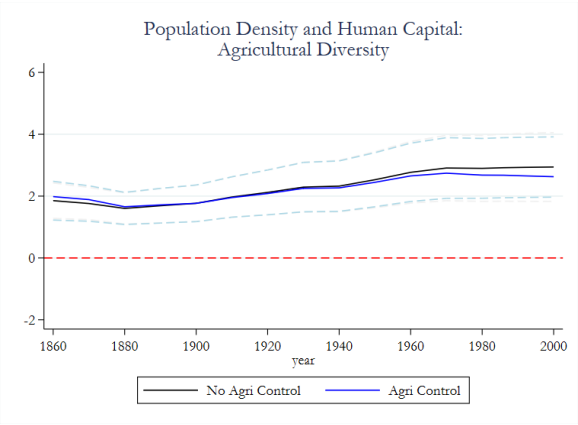
Figure E1 shows the resulting coefficient paths. Copyrighting retains positive historical associations conditional on agricultural diversity. The agricultural-diversity relationship also changes when copyrighting enters. These estimates indicate that recorded intellectual production contains information about local development beyond agricultural diversity; their IV interpretation requires the exclusion restrictions for both instruments.



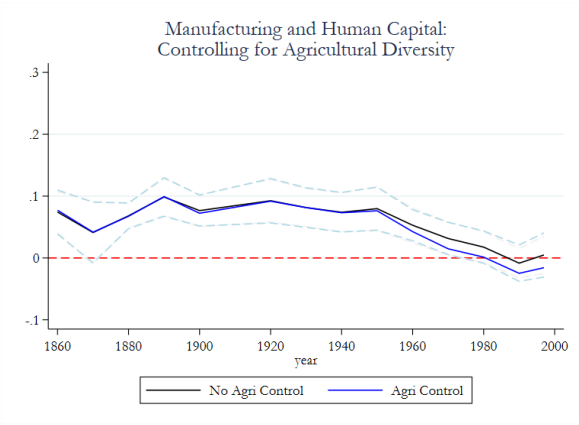
(a) Diversity: population density



(b) Diversity: manufacturing share



(c) Copyrighting: population density



(d) Copyrighting: manufacturing share

Figure E1. Copyrighting and agricultural diversity in joint specifications

Notes: Reported IV coefficient paths from the supplementary joint analysis. Panels (a) and (b) show agricultural-diversity coefficients conditional on copyrighting; panels (c) and (d) show copyright coefficients conditional on agricultural diversity. The agricultural-diversity instrument follows Fiszbein (2022). These results are distinct from the baseline specifications in the main text.

E.4 Religious composition and copyrighting

The comparison of religious composition and copyrighting also speaks to research on diversity and economic performance. Studies of ethnic and religious diversity examine their connections to conflict and development (Montalvo and Reynal-Querol, 2003, 2005; Alesina and La Ferrara, 2005), while Ager and Brückner (2013) studies cultural diversity in the American context of mass migration. The church-based indices used here describe the local distribution of religious organizations. They supply descriptive evidence on the setting of written production, distinct from the birthplace composition used in the instrument. This distinction keeps the religious correlations separate from the interpretation of the manufacturing results.

The church data permit a descriptive comparison of written production and local religious composition. Following Montalvo and Reynal-Querol (2003), let n_{ij} be the number of churches of denomination j in county i, let $N_i = \sum_j n_{ij}$, and let $\pi_{ij} = n_{ij}/N_i$. Denominational fractional-

ization and polarization are

$$\mathrm{FRAC}_i = 1 - \sum_{j=1}^{J} \pi_{ij}^2, \tag{E2}$$

$$\mathrm{POL}_i = 1 - \sum_{j=1}^{J} \left(\frac{0.5 - \pi_{ij}}{0.5} \right)^2 \pi_{ij}. \tag{E3}$$

Fractionalization is the probability that two randomly drawn churches belong to different denominations. Polarization is greatest when two denominations each account for half the churches. The indices use 1850 church counts from Haines and ICPSR (2010) and therefore describe counties during the copyright registration window.

Table E3 includes both indices in regressions of log copyright registrations over 1840–1860. Fractionalization is positively associated with copyrighting and polarization negatively associated with it. Both remain statistically significant throughout the control sequence; their fully controlled coefficients are 0.268 and −0.215, respectively. These contemporaneous partial correlations describe the religious settings in which written production was more prevalent; they do not establish a direction of causation.

Table E3. Religious fractionalization, polarization, and county copyrighting

	(1)	(2)	(3)	(4)	(5)
Fractionalization	0.582	0.558	0.523	0.523	0.268
	(7.30)	(8.05)	(7.92)	(7.92)	(5.59)
Polarization	−0.505	−0.476	−0.445	−0.445	−0.215
	(−6.38)	(−6.73)	(−6.23)	(−6.23)	(−5.14)
Observations	1511	1506	1505	1505	1504
State fixed effects	Yes	Yes	Yes	Yes	Yes
Latitude–longitude polynomial	Yes	Yes	Yes	Yes	Yes
Geographic controls	No	Yes	Yes	Yes	Yes
Human-capital controls	No	No	Yes	Yes	Yes
Social controls	No	No	No	No	Yes

Notes: OLS estimates. The dependent variable is $\log(1 + C_i)$, with C_i denoting county copyright registrations over 1840–1860. Both religious-composition indices enter each specification. Parentheses contain t statistics based on Conley standard errors with a 100 km cutoff. Control groups are defined in Table B1.

F An illustrative model of skill accumulation and cultural transmission

F.1 Purpose and environment

This appendix sets out a mechanism through which the composition of a community can influence skill accumulation across generations. It combines the intergenerational transmission of cultural types in Bisin and Verdier (2001) with the education and fertility decisions central to unified growth models (Galor, 2011). The distinction is that households differ in their ability to use skills and in their preferences for acquiring them. A child's future type depends both on parental effort and on the composition of the surrounding population. The mechanism can therefore connect initial settlement composition to subsequent human-capital decisions without requiring that every written work directly advance production technology.

There are two types, traditional (T) and modern (M). These names denote assumed preferences and production opportunities, rather than empirical categories assigned to authors. Households live for one period and choose consumption, fertility, education, and socialization effort. Production takes place at the household level, without outside hiring. Traditional households use labor and land; modern households can also use skills in production. The state variables are inherited skills, S, and the traditional population share, q. Primes refer to the next generation. Uppercase subscripts distinguish types from calendar time.

F.2 Household decisions

Let c_i denote consumption, n_i the number of children, e_i educational investment, and τ_i^q direct socialization effort for type i. Socialization effort controls the probability that a child acquires the parent's type. The traditional household solves

$$\begin{aligned} V_T(S,q) = \max_{c_T,n_T,e_T,\tau_T^q} \Big\{ & u^T(c_T,n_T) \\ & + \beta\pi_{TT}(\tau_T^q,q)V_T(S_T',q') \\ & + \beta\pi_{TM}(\tau_T^q,q)V_M(S_T',q') \Big\}. \end{aligned} \tag{F1}$$

The modern household solves

$$\begin{aligned} V_M(S,q) = \max_{c_M,n_M,e_M,\tau_M^q} \Big\{ & u^M(c_M,n_M,S) \\ & + \beta\pi_{MM}(\tau_M^q,q)V_M(S_M',q') \\ & + \beta\pi_{MT}(\tau_M^q,q)V_T(S_M',q') \Big\}. \end{aligned} \tag{F2}$$

The discount factor β weights the continuation value associated with the child's skills and cultural type. Traditional households value consumption and children; modern households also value skills directly. The probabilities in the continuation terms are defined below. In the skill law, δ denotes the retained fraction of inherited skills, so that $1-\delta$ is the depreciation rate.

Consumption is financed by household labor income:

$$c_T = w_T l_T, \qquad c_M = w_M(S) l_M. \tag{F3}$$

Traditional earnings per unit of labor do not depend on skills, whereas modern earnings do.[2] Each household has one unit of time. Raising a child takes a fixed amount $\bar{\tau}$, in addition to socialization and education time:

$$l_i + n_i\left[\bar{\tau} + \tau_i^q + \tau^e(S)e_i\right] = 1, \qquad i \in \{T,M\}. \tag{F4}$$

The function $\tau^e(S)$ converts educational investment into its parental time cost. Skills passed to the next generation satisfy

$$S_i' = f(S,e_i) + \delta S. \tag{F5}$$

[2]The traditional income opportunity can be motivated by the labor-and-land technology $F(L,X) = (AX)^\alpha L^{1-\alpha}$, where L is labor, X land, and A productivity. This production function is the proposed background for w_T; the framework does not specify a complete system of land allocations or factor-market clearing conditions.

The skill-production function is increasing and concave. The specific functional forms used for illustration are

$$\begin{aligned} u^M(c,n,S) &= \chi \ln(c-\bar{c}) + \ln(S) + \ln(n), \\ u^T(c,n) &= \chi \ln(c-\bar{c}) + \ln(n), \\ \tau^e(S) &= S^{-1}, \\ f(S,e) &= S^{\nu} e^{1-\nu}. \end{aligned} \tag{F6}$$

Here $\bar{c}$ is subsistence consumption and ν controls the contribution of parental skills to children's skill development. With $\chi > 0$ and $0 < \nu < 1$, utility is increasing and strictly concave on its domain, while the skill technology is jointly concave and strictly concave in either input holding the other fixed. The skill technology has constant returns in its two inputs. Feasible choices have $c_i > \bar{c}$, $n_i > 0$, $e_i \geq 0$, $l_i \geq 0$, and $\tau_i^q \in [0,1]$, with positive inherited skills. Higher inherited skills both contribute to children's skills and reduce the time cost of education.

F.3 Cultural transmission and population composition

A parent succeeds in direct socialization with probability τ_i^q. If direct socialization fails, the child's type is drawn from the community composition. For a traditional parent,

$$\begin{aligned} \pi_{TT}(\tau_T^q, q) &= \tau_T^q + (1-\tau_T^q)q, \\ \pi_{TM}(\tau_T^q, q) &= (1-\tau_T^q)(1-q). \end{aligned} \tag{F7}$$

For a modern parent, the corresponding probabilities are

$$\begin{aligned} \pi_{MM}(\tau_M^q, q) &= \tau_M^q + (1-\tau_M^q)(1-q), \\ \pi_{MT}(\tau_M^q, q) &= (1-\tau_M^q)q. \end{aligned} \tag{F8}$$

Thus, at a given parental effort, a child is more likely to become modern in a population with a larger modern share, regardless of the parent's type. This is the source of the local socialization effect.

The next generation's traditional share weights these probabilities by the number of children born to each type:

$$\begin{aligned} q' &= \frac{q n_T \pi_{TT} + (1-q) n_M \pi_{MT}}{q n_T + (1-q) n_M} \\ &= \frac{q n_T [\tau_T^q + (1-\tau_T^q)q] + (1-q) n_M (1-\tau_M^q) q}{q n_T + (1-q) n_M}. \end{aligned} \tag{F9}$$

Composition is therefore endogenous to fertility as well as direct socialization. An initial difference in q changes the social environment facing parents; their responses in turn change the environment inherited by their children.

F.4 Interpretation and scope

The mechanism operates through the allocation of parental time. When modern types are scarce, a modern parent faces a greater chance that unsuccessful direct socialization produces a traditional child. Effort devoted to preserving the parent's type uses time that could otherwise support education, consumption, or fertility. A larger modern population can relax this pressure.

Traditional parents can also have an incentive to educate children, despite earning no direct production return from their own skills: their children may become modern through socialization and then use those skills. These are conditional incentives; their net effect depends on the preferences, earnings, and policy responses in the household problems.

The dynamics in equation (F9) illustrate how communities with different initial population compositions may follow different paths of skill accumulation even when households have identical preferences and technologies within each type. The framework does not assign these types to individuals in the census or identify their transition probabilities from the copyright data.

The equations characterize household incentives and population-composition accounting; they do not close a quantitative transition model for the distribution of skills. The framework offers an interpretation of the long-run relationship between settlement composition and economic development. It has not been calibrated to the copyright data, and the empirical analysis does not separately identify cultural transmission.

G An analytical illustration of incomplete measurement

Copyrighting records only part of intellectual activity. The following example makes precise how incomplete measurement can separate OLS and IV coefficients under particular assumptions. The argument is analytical; its parameters are not estimated from the county data.

Let the outcome depend on total knowledge, x, while the econometrician observes only $\widetilde{x}$:

$$\begin{aligned} y &= a + bx + \varepsilon, \\ x &= d\widetilde{x} + (1-d)f, \qquad 0 < d < 1, \\ z &= \widetilde{x} + f + \eta. \end{aligned} \tag{G1}$$

Here f is an unobserved component of knowledge and η is noise in the instrument. To complete the example, impose the additional normalization $\mathrm{Var}(\widetilde{x}) = \mathrm{Var}(f) = 1$, with both components mean zero and jointly normal, and let $\mathrm{Cov}(\widetilde{x}, f) = \rho$. Under this normalization, ρ is also their correlation. Assume that ε has mean zero, finite variance, and is independent of $(\widetilde{x}, f, \eta)$, and that $\eta \sim N(0, \sigma_e)$ is independent of $(\widetilde{x}, f)$; σ_e denotes the noise variance. Equal component variances and these independence conditions are assumptions for the illustration, not properties established for the historical data.

The population OLS coefficient from regressing y on $\widetilde{x}$ and an intercept is

$$\begin{aligned} \beta^{\mathrm{OLS}} &= \frac{\mathrm{Cov}(\widetilde{x}, y)}{\mathrm{Var}(\widetilde{x})} \\ &= b\big[d + (1-d)\rho\big]. \end{aligned} \tag{G2}$$

For positive b and $0 \le \rho < 1$, this coefficient is below b. The gap is larger when the observed component receives less weight in total knowledge or is less correlated with the unobserved component. This is an incomplete-proxy model: attenuation follows from its specific aggregation and covariance structure, rather than from the presence of measurement error alone.

Instrumenting $\widetilde{x}$ with z gives the population estimand

$$\begin{aligned}\beta^{\mathrm{IV}} &= \frac{\mathrm{Cov}(z,y)}{\mathrm{Cov}(z,\widetilde{x})} \\ &= b\frac{d(1+\rho)+(1-d)(1+\rho)}{1+\rho} = b, \qquad \rho > -1. \end{aligned} \tag{G3}$$

The equality depends on the instrument having the same covariance with the observed and unobserved components. The equal-variance normalization ensures this property for the specified sum in equation (G1). It is not a general consequence of using an instrument correlated with some omitted knowledge. When $\rho = -1$, the relevant covariance is zero and the ratio is undefined.

Independent instrument noise leaves these population covariances unchanged but increases $\mathrm{Var}(z)$. It therefore reduces the explanatory power of the first stage and the information available for estimating the IV coefficient. In particular,

$$\mathrm{Corr}(z,\widetilde{x})^2 = \frac{(1+\rho)^2}{2(1+\rho)+\sigma_e}. \tag{G4}$$

The suggested values $b = 0.7$, $\rho \in \{0.9, 0.1\}$, and $\sigma_e = 1$ are illustrative parameters. They compare a highly informative observed component with one less closely related to the unobserved component; they are neither a calibration nor estimates from the county data.

The calculation illustrates how incomplete measurement can produce larger IV than OLS estimates under the stated assumptions. It does not establish that the difference in the county estimates is due to measurement error. In particular, the IV estimates in the main analysis also use a different control set, and their interpretation requires the settlement-based exclusion restriction.